LUDOVIC MOMPELAT - UNIVERSITY OF MIAMI

FEB 21 2025 – 3RD ANNUAL CARIBBEAN DIGITAL HUMANITIES VIRTUAL LECTURE

LINGUISTIC CHALLENGES AND AI INNOVATIONS FOR HAITIAN CREOLE: OVERCOMING LANGUAGE BARRIERS FOR HEALTHCARE

ADDRESSING LANGUAGE BARRIERS IN HEALTHCARE

- **M**any Low English Proficiency (LEP) patients in the U.S. struggle to access healthcare due to communication challenges.
- **L**anguage barriers negatively impact health outcomes. (Pérez-Escamilla et al., 2010)
- **L**imited access to accurate medical communication increases risks of medical errors and poor patient adherence to treatment.
- **T**he Haitian diaspora in the U.S. is one of the most linguistically vulnerable populations.
- **M**iami has one of the largest Haitian communities.

FROM *HABLAMOS JUNTOS* INITIATIVE…

- **Hablamos Juntos** launched in 2002 aimed at improving language access for Spanish-speaking Latinos in U.S. healthcare (Moreno & Morales, 2010).
- **F**ocus on standardizing interpreter services.
- Despite its success, a major challenge remains:
  "*The absence of universally available language services is a national healthcare system failure, the burden of which is suffered by patients with limited English proficiency and their healthcare providers.*" *Partida (2007), Director of the National Program Office for Hablamos Juntos*
- Partida (2007) calls for a shift toward AI & technology-based solutions, stating: "*sustained investments in population-based solutions that leverage the power of computers and communication technology can lead to solutions that can reach across boundaries of responsibility to enable large and small healthcare provider organizations to serve patients of many languages.*"

Hablamos Juntos
We speak together.

TO… *NOU PALE ANSANM*

This initiative stems from similar fundamental issue:

- **A** lack of readily available, accurate translation tools
- **S**uccessful medical communication remains a challenge.

What's the Problem with Existing Solutions?

No high-quality Haitian Creole language tools

- Unable to fully meet healthcare communication needs.

Shortage of trained HC medical interpreters

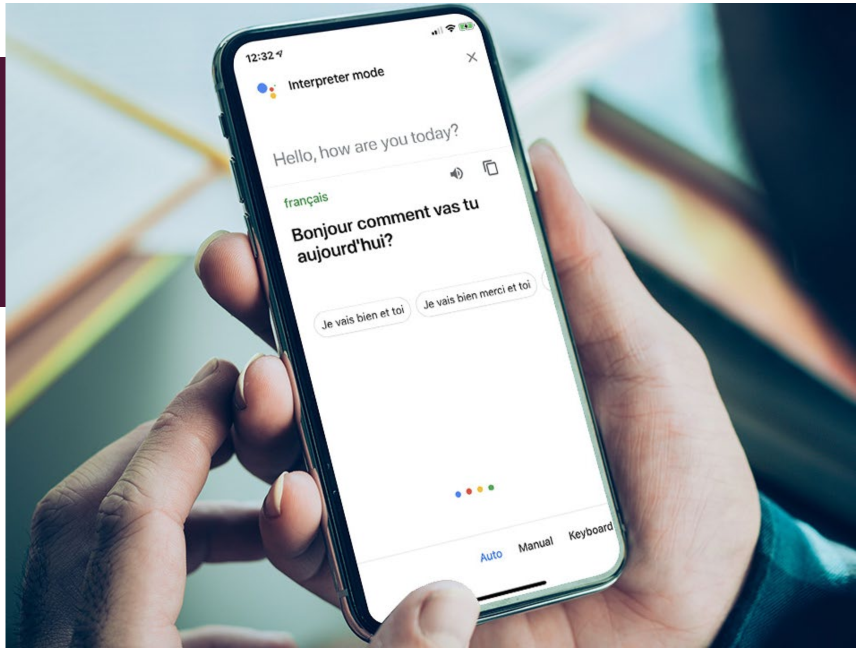
- Leads to reliance on underqualified individuals.

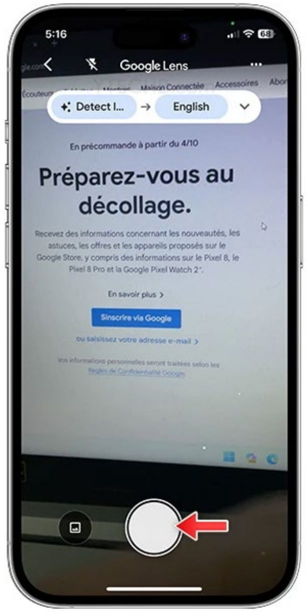
Over-reliance on family members for translation

- Can compromise accuracy, privacy, and patient autonomy.

PROJECT OVERVIEW

- This long-term initiative carried by Dr. Tézil and myself is the following :
  - Propose a reliable, scalable and cost-effective translation tool for Haitian Creole for Patient-Provider communication.
- Potential Applications:
  - Real-time interpretation for medical interactions.
  - Prescription translation & patient instructions.
  - Improved healthcare accessibility for Haitian Creole-speaking LEP patients.
- Why AI?
  - Available human interpreters are absent in many healthcare settings.
  - **A**I complements interpreters' work, ensuring continuous medical translation availability.
  - Recent advances in Natural Language Processing (NLP) & Machine Learning (ML) allow for more accurate, domain-specific translations.





WHAT HAITIAN CREOLE?

French-based Creole spoken by millions

Exhibits significant linguistic variation from more basilectal/Creole-like to mesolectal/French-influenced varieties (Fattier-Thomas 1984, Valdman 2015)

- **Kreyòl Swa (KS):** Spoken by bilingual Haitian speakers, more influenced by French.
  - Becoming more widespread, even among monolingual speakers (Tézil, 2024).
- **Kreyòl Rèk (KR):** Spoken by monolingual Haitian Creole speakers, less influenced by French
- Haitian Creole norm based on KR (esp. monolingual speech from Port-au-Prince).
- Diaspora communities speak Kreyòl Swa and possess varying degrees of knowledge of French.

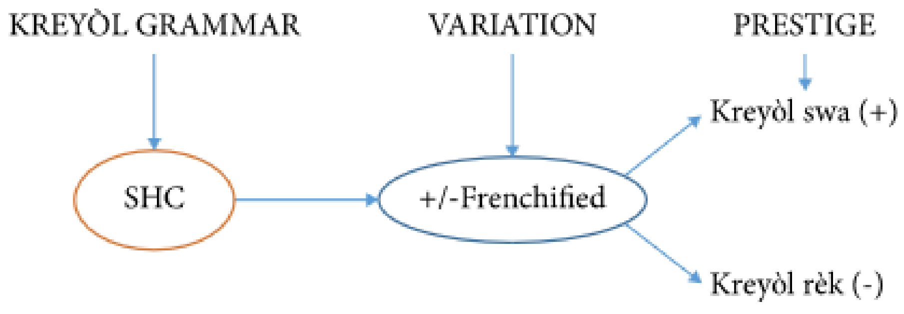


Figure 6. Representation of KR and KS

Tézil 2022 p.310

1. Les douleurs musculaires avaient diminué [FR]
   DET pains muscular had diminished
2. doulè nan misk la te bese [GT HC]
   pain in muscle DET.SG ANT lower
3. doulèur muskulèr yo te diminue [NS HC]
   pain muscular DET.PL ANT diminish

« *Muscle pain had diminished* » Tézil 2025, p.20

A STARTING POINT…

- Pre-existing available resources for HC
  - Carnegie Mellon University initiative

[HC] e nou dwe kontwòle tou kolestewòl ak tansyon ou
[ENG] and also needs to be checked your cholesterol blood pressure

  - Universal Dependencies treebank for Haitian Creole (Jagodzińska et al.)
  - Leipzig Corpora Collection

```
# sent_id = 01__2
# text = Jou sa lapli te kòmanse tonbe depi nan maten.
# Status = SILVER
# text_fr = Ce jour là, il commençait à pleuvoir dans la matinée.
1	Jou	jou	NOUN	_	_	5	obl:mod	_	Gloss=jour
2	sa	sa	DET	_	PronType=Dem	1	dep	_	Gloss=cet
3	lapli	lapli	NOUN	_	_	5	nsubj	_	Gloss=pluie
4	te	te	AUX	_	Tense=Past	5	aux	_	Gloss=PAST
5	kòmanse	kòmanse	VERB	_	_	0	root	_	Gloss=commencer
6	tonbe	tonbe	VERB	_	_	5	ccomp	_	Gloss=tomber
7	depi	depi	ADV	_	_	5	advmod	_	Gloss=depuis
8	nan	nan	ADP	_	_	9	case	_	Gloss=dans
9	maten	maten	NOUN	_	_	7	obl	_	Gloss=matin|SpaceAfter=No
10	.	.	PUNCT	_	_	5	punct	_	_
```

Akademi Kreyòl Ayisyen

3 lang

Atik Diskisyon Li Modifye Modifye kòd Gade istorik Bwat zouti

Depi Wikipedya, ansiklopedi lib

Akademi Kreyòl Ayisyen, se enstans nan peyi Ayiti ki la pou regile lang kreyòl la.[1] Konstitisyon 1987 nan atik 213 rekonèt egzistans Akademi an. Li konpoze de jiska 55 savan anba lidèchip Jean Pauris Jean-Baptiste.[2]

Istwa li [modifye | modifye kòd]

Lang kreyòl ayisyen an pa gen okenn règleman jiskaske ane 1940 yo, lè ansyen prezidan ayisyen, Élie Lescot te fè tantativ nan estandadize lang lan. Li te gen yon otograf ofisyèl pa fen ane 1970 yo, ak li te elve pou yo te lang ofisyèl la nan Konstitisyon an Ayiti. Lang la toujou te manke yon akademi regle evolisyon li jiskaske apeprè 25 ane pita.

DATA SOURCES FOR HAITIAN CREOLE

- Pre-existing available resources for HC
  - Spanish & Haitian Creole Medical Terminology: Medical Translation Pocket Booklet (Azolin et al. 2023)
  - Patient-Provider Bilingual Tools Haitian Creole and English (USSAAC 2023)

Commands		
English	**Creole**	**Spanish**
Sit down	Chita	Siéntese
Stand up	Kanpe	Levántse
Open your eyes	Ouvri Je ou	Abra los ojos
Close your eyes	Fèmen je ou	Cierre los ojos
Turn your head	Vire tèt ou	Vire la cabeza
Open your mouth	Louvri bouch ou	Abra la boca
Stick out your tongue	Mete lang ou deyò	Saque la lengua
Lie down	Kouche	Acuéstese
Raise your hand	Leve men ou	Suba la mano
Squeeze my hand	Peze men mwen	Apriete mi mano
Raise your Legs	Leve janm ou	Suba la pierna
Take a deep breath	Pran yon gwo souf	Respire profundo

SUCTION ASPIRASYON	WHAT'S MY STATUS? KISA ETA MWEN YE?	CALL MY FAMILY RELE FANMI M	LIGHTS ON/OFF LIMYÈ YO LIMEN/ETENN
TROUBLE BREATHING TWOUB RESPIRASYON	PAIN DOULÈ	MEDICINE MEDIKAMAN	HOT COLD CHO / FRÈT
BATHROOM TWALÈT	REPOSITION REPOZISYONE	MOUTH CARE SWEN POU BOUCH	LETTER BOARD TABLO LÈT YO
MAYBE - PETÈT	DON'T KNOW – PA KONNEN	LATER - PITA	

WHAT IS MY PROGNOSIS? KISA PREVIZYON MWEN YE?	WHAT ARE MY OPTIONS? KISA OPSYON MWEN YO YE?	WILL I GET BETTER? M AP FÈ MIYÒ?	AM I GOING TO DIE? ÈSKE MWEN PRAL MOURI?
WHAT WILL HAPPEN NEXT? KISA KI PRAL RIVE ANSUIT?	WILL I HAVE PAIN? ÈSKE MWEN PRAL GEN DOULÈ?	I WANT TO DISCUSS MY DECISIONS MWEN VLE DISKITE SOU DESIZYON MWEN YO.	I WANT MY FAMILY TO DECIDE MWEN VLE FANMI MWEN DESIDE
WHEN WILL I COME OFF THE VENTILATOR? KILÈ MWEN AP SOT NAN RESPIRATÈ AN?	WHAT HAPPENS IF I AM TAKEN OFF THE VENTILATOR? KISA KI PRAL RIVE SI YO RETIRE M SOU RESPIRATÈ AN?	I AM NOT READY TO MAKE A DECISION MWEN PA PARE POU PRAN DESIZYON	I HAVE ANOTHER QUESTION MWEN GEN YON LÒT KESYON.
MAYBE - PETÈT	DON'T KNOW – PA KONNEN	LATER - PITA	

LIMITATIONS

Datasets	Limitations
Carnegie Mellon University initiative	Approximative translations – require manual annotation and validation
Universal Dependencies treebank for Haitian Creole (Jagodzińska et al.)	Highly linguistically informative – limited quantity and genre represented
Leipzig Corpora Collection	High token quantity – unannotated data and genre-specific
Spanish & Haitian Creole Medical Terminology (Azolin et al. 2023)	Domain-specific – limited quantity and limited representation of actual language practice and use
Patient-Provider Bilingual Tools (USSAAC 2023)	Domain-specific – limited quantity and limited representation of actual language practice and use

- Need for Data validation and augmentation – work in progress, in collaboration with other researchers

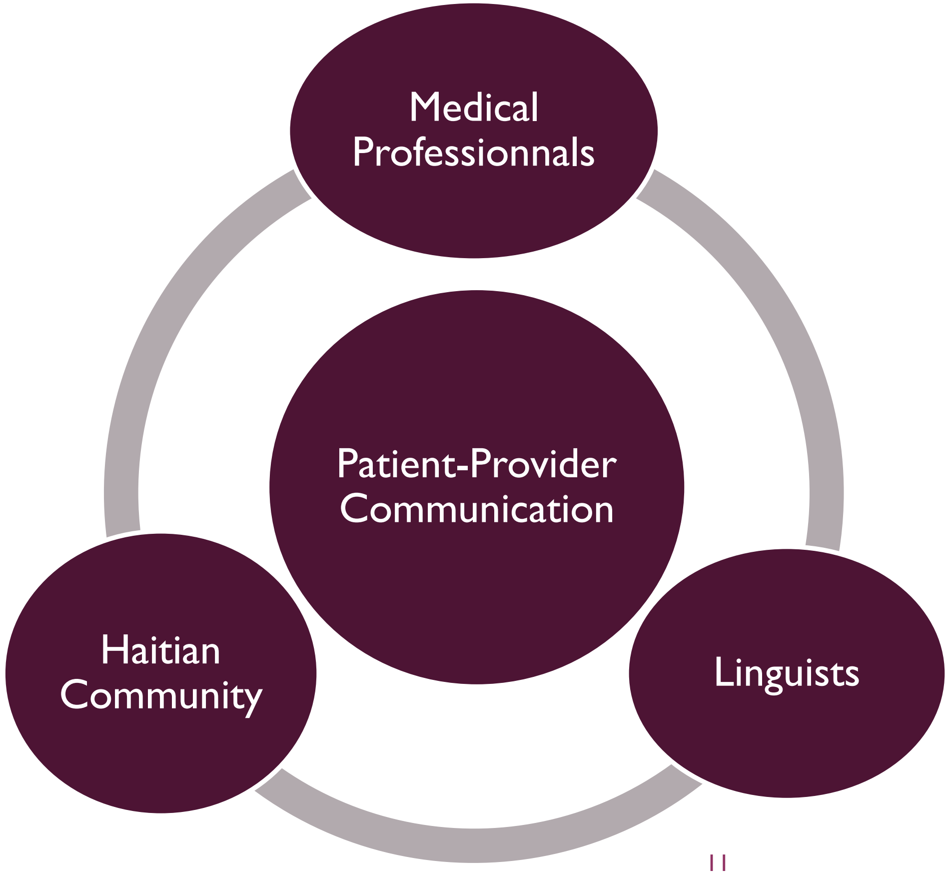
COMMUNITY ENGAGEMENT

- Community engagement for data augmentation and validation
  - **M**edical professionals in Miami like Dr. Marie Denise Gervais
    - **S**eek to expand our collaborations with medical
  - **I**n touch with members of the HC-speaking communities in Miami and other parts of the US
  - **P**ossible collaborations with more linguists and Creolists in the field
- Follow advances in NLP for Creole
  - **C**reoleval project (Lent et al. 2022, Lent et al. 2024)
  - **M**ultilingual language models – mBERT (Devlin et al. 2018), XLM-R (Conneau et al. 2019), mT5 (Xue et al. 2020), and M2M-100 (Fan et al. 2020)

[Translation tool](#) for [Haitian Creole](#) for **[Patient-Provider communication](#)**

WHAT PATIENT-PROVIDER COMMUNICATION?

- Lack of trained interpreters forces reliance on family members or informal translation, risking miscommunication.
- Improve the quality of care via reliable medical communication and translation tools
  - Requires precision to prevent misdiagnoses & ensure patient safety.
  - **M**ust drastically reduce the risk of translation errors
- Finding the right balance
  - **M**edical terminology must be accurately translated.
  - **L**inguistic/cultural sensitivity must be a focus.



WHY A LINGUISTICALLY-INFORMED APPROACH?

- In absence of large datasets, it can help inform strategies for adapting pre-existing AI language models.
- Ensures that low-resource languages like (HC) are accurately represented in multilingual models & LLMs.
- The MT models must be sensitive to linguistic variation
  - **B**etween KS and KR
  - **B**etween HC and French/English/Spanish
- In both cases, models must be able to decode AND encode linguistic variation for appropriate communication

What to consider?
• HC-French Code-Switching & Mixed Speech Patterns.
• Contrasting KS & Standard Haitian Creole.
• Linguistic Attitudes & Speaker Expectations.

WHAT TRANSLATION TOOL?

Machine translation (MT) : use of AI to translate text or speech from one language to another.

Rule-based MT → Statistical MT (SMT) → Neural MT (NMT)

MT for low-resource languages often fails (Ranathunga et al. 2023) :

- **Parallel data scarcity** : require millions/billions of parallel pairs, which low-resource languages like Haitian Creole lack.
- **Poor Domain adaptation** : General models (e.g., Google Translate) underperform in healthcare contexts.

- The Problem with General-Purpose Translators
  - No guarantees of translation accuracy, especially for complex medical terms.
  - Fail to capture cultural nuances and medical literacy levels.
  - Jeopardize patient-provider communication

Brewster et al. 2024	Translators	GoogleTranslate	ChatGPT
Adequacy	4.5	4.0	3.9
Fluency	3.9	3.6	3.4
Meaning	4.0	3.7	3.6
Severity	4.5	4.0	3.8

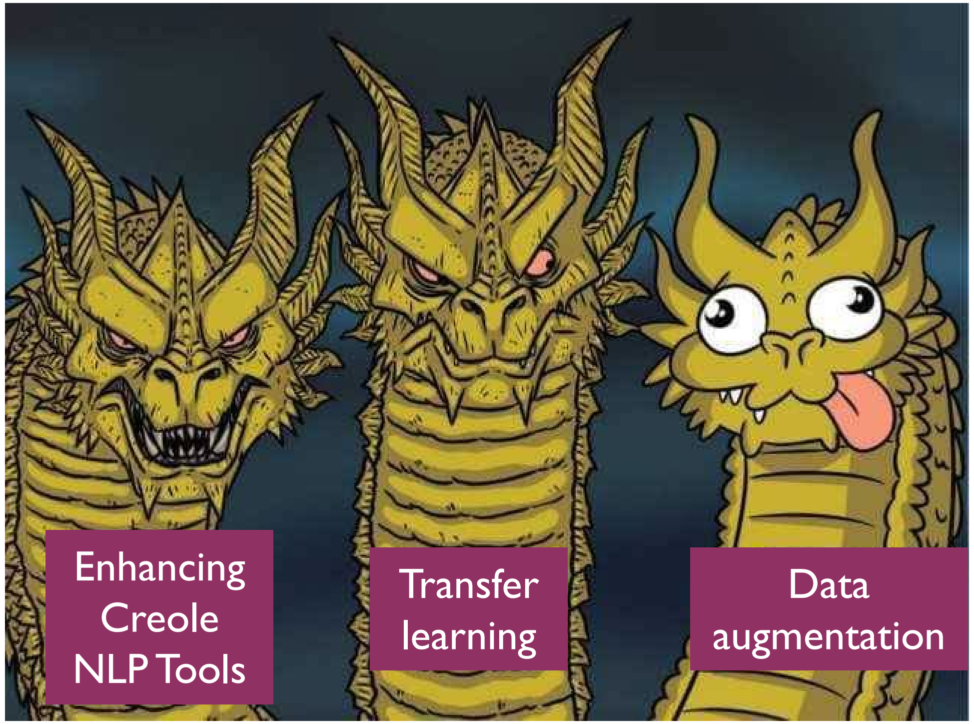
- Human translators consistently outperform Google Translate & ChatGPT in adequacy, fluency, meaning, and severity ratings (out of 5).
- ChatGPT & Google Translate show notable weaknesses in medical translation accuracy, reinforcing the need for domain-specific AI solutions.

THE GAME PLAN

Strategy : A hybrid linguistically-informed approach to MT

- Provide our model with more linguistic information instead.
  - Linguistically informed models can fill data gaps and improve translation accuracy.
  - AI models must integrate knowledge of Haitian Creole's linguistic structure & variation to be truly effective.

Proposal : Three-part strategy
Enhancing Creole NLP Tools : Develop parsers, UD datasets, POS taggers.
Transfer Learning: Leverage French-based models to improve Creole translation.
Synthetic Data Augmentation: Generate parallel KS/KR & English-Creole datasets.



SYNTHETIC DATA AUGMENTATION

Expanding Creole Data for KS/KR & Healthcare Translation synthetically

- With synthetic data we can leverage the lack of resources and produce more culturally and linguistically accurate representations of Haitian Creole for our model.

Step 0: Baseline Model Setup

- **T**est and fine-tune existing LLMs for KS/KR distinction, and domain-specific HC

Step 1: Data Augmentation Through Synthetic Generation

- **L**everage LLMs to create parallel HC-English datasets and KS/KR medical translation datasets.

Step 2: Human Validation & Annotation

- **M**anual annotation and review of synthetic data to ensure linguistic and domain accuracy.

Step 3: Reinforcement Learning for LLMs - Semi-automatic iterative process:

- Validated synthetic data is fed back into the model for continuous improvement.
- Human feedback fine-tunes performance, ensuring dialect accuracy.
- NLP tools assist in evaluating and structuring synthetic text.

BUILDING BETTER NLP TOOLS FOR CREOLE

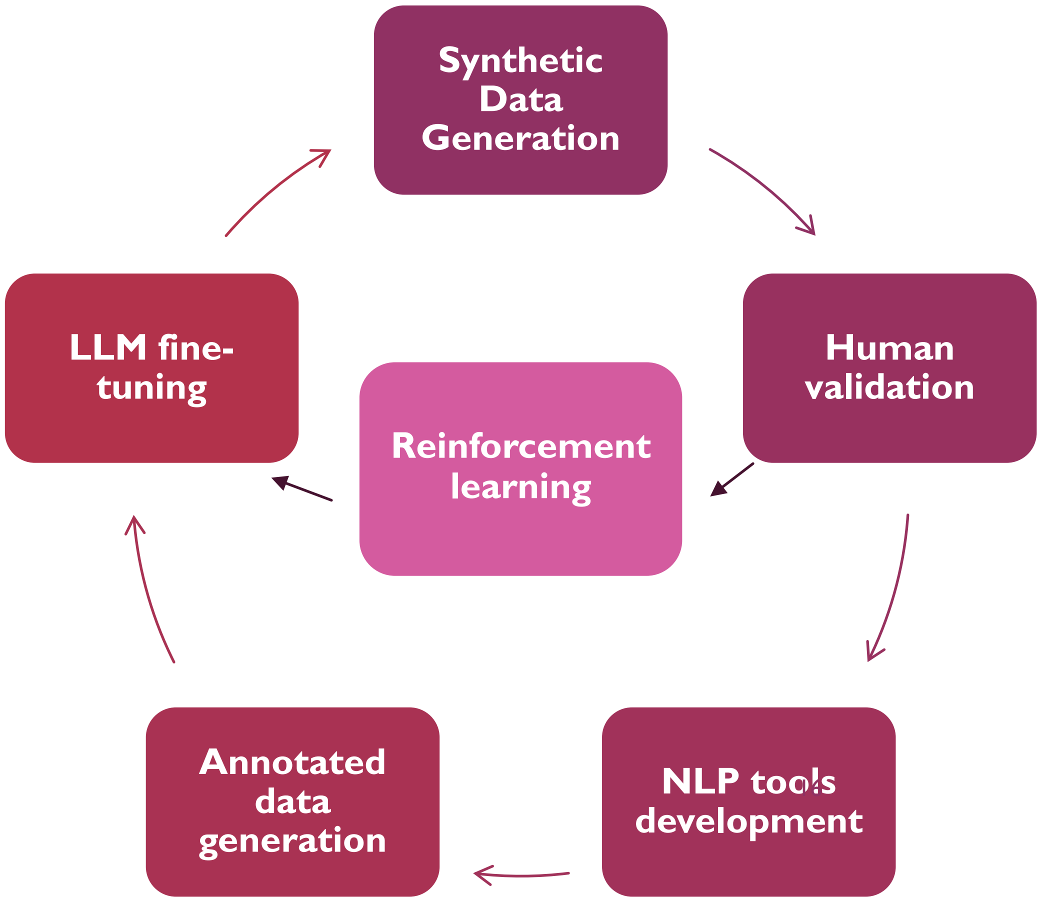
Creating better NLP tools for HC can enhance the performance of the LLMs on top of our reinforcement learning approach.

- Enhance Universal Dependencies (UD) datasets for Creole.
- Improve Creole POS tagging & syntax models.
- Develop parsing tools to create more annotated data

Cool, but how?

- Pre-existing datasets + additional manual annotations
- Transfer Learning & Multitask Learning (Mompelat et al., 2022)
  - Use high-resource languages like French as a baseline for training Creole models.
  - Fine-tune models on Haitian Creole corpora to improve adaptability.

A continuous feedback loop



CONCLUSION

- So now we know how and why to build a reliable, scalable and cost-effective translation tool for Haitian Creole for Patient-Provider communication in Miami.
- What's the end goal again?
  - Deploy an automatic translation system for patient-provider communication in Miami healthcare settings.
  - Ensure accuracy in real-world contexts by integrating variation-sensitive MT models.
  - Design a tool that is scalable and adaptable for integration in the healthcare system.

BROADER IMPLICATIONS FOR NLP & LINGUISTICS

- For Haitian Creole and Code-Switching research :
  - Miami's high CS environment makes it a rich site for NLP research on CS phenomena.
  - Advancing NLP capabilities to process mixed Creole-French-English structures fills a major research gap.
  - Research in leveraging Creole-lexifier similarities to improve NLP models across Creole languages
- For Creolistics & Linguistic Continuum Research :
  - This initiative contributes to ongoing debates in Creolistics regarding:
    - The Creole-lexifier relationship and its computational modeling.
    - Linguistic continua in Creoles and their implications for MT & NLP models.
  - Forces NLP & Linguistics to rethink traditional modeling strategies beyond rigid language boundaries.
- For Machine Translation and LLM research :
  - Developing MT & LLM agents capable of handling code-switching remains an open challenge in NLP.
  - This research paves the way for NLP innovations in low-resource and multilingual contexts.

WHAT'S NEXT?

Short-term plan : Laying the Foundations

- Leverage pre-existing language models & LLMs to:
  - **A**ugment open-source Creole datasets.
  - Continue experimenting with other Creole languages to assess cross-linguistic transfer effects.
- Finalize Pilot Research
  - Initial zero-shot MT experiments show poor performance.
  - Evaluate & refine models iteratively.
  - Prototype a working translation system to test in pilot healthcare settings.

Long-Term Vision: Sustainable system

- Seek funding for computationally expensive tasks, including:
  - Scaling reinforcement learning for Creole
  - Fine-tuning translation models for specialized healthcare domains.
  - **B**uilding robust, open-source Creole NLP tools.
- Community & Medical Collaboration for Real-World Impact
  - Develop datasets representative of real-life Creole usage.
  - Ensure cultural & ethical sensitivity via community engagement.
  - Deploy & evaluate the translation tool in actual healthcare settings, iterating based on real-world feedback.

THANK YOU

REFERENCES

- Azolin, Christelle R. Stephanie A. Martinez, Veronica Alvarez, Tristan Vulcain "Spanish and Haitian Creole Medical Terminology: Medical Translation Pocket Booklet." FIU Digital Commons, https://dx.doi.org/10.25148/hwcom.010894
- Brewster, Ryan CL, Priscilla Gonzalez, Rohan Khazanchi, Alex Butler, Raquel Selcer, Derrick Chu, Barbara Pontes Aires, Marcella Luercio, and Jonathan D. Hron. "Performance of ChatGPT and Google Translate for Pediatric Discharge Instruction Translation." In *Pediatrics*. 2024. e2023065573.
- Fattier-Thomas, Dominique. "De la variété rèk à la variété swa: Pratiques vivantes de la langue en Haïti." *Conjonction161* 162 (1984): 39-51.
- Mompelat, Ludovic, Daniel Dakota, and Sandra Kübler. "How to Parse a Creole: When Martinican Creole Meets French." In *Proceedings of the 29th International Conference on Computational Linguistics*, pp. 4397-4406. 2022.
- Moreno, Gerardo, and Leo S. Morales. "Hablamos Juntos (Together We Speak): interpreters, provider communication, and satisfaction with care." *Journal of general internal medicine* 25 (2010): 1282-1288.
- Partida, Yolanda. "Addressing language barriers: Building response capacity for a changing nation." *Journal of General Internal Medicine* 22 (2007): 347-349.
- Pérez-Escamilla, Rafael, Jonathan Garcia, and David Song. "Health care access among Hispanic immigrants:¿ Alguien está escuchando?[Is anybody listening?]." *NAPA bulletin* 34, no. 1 (2010): 47-67.
- Ranathunga, Surangika, En-Shiun Annie Lee, Marjana Prifti Skenduli, Ravi Shekhar, Mehreen Alam, and Rishemjit Kaur. "Neural machine translation for low-resource languages: A survey." *ACM Computing Surveys* 55, no. 11 (2023): 1-37.
- Valdman, Albert. *Haitian Creole: structure, variation, status, origin*. Sheffield: Equinox, 2015.
- Tezil, David. "On the influence of Kreyòl swa: Evidence from the nasalization of the Haitian Creole determiner/la/in non-nasal environments." *Journal of Pidgin and Creole Languages* 37, no. 2 (2022): 291-320.
- Tézil, David. "Sociolinguistic challenges and new perspectives on determining French speakers in Creole communities: the case of Haiti." *International Journal of the Sociology of Language* 2024, no. 288 (2024): 177-207.